

大模型 + 数据资产变现 RAG驱动企业智能化实践案例

演讲人：黄佳

新加坡科研局 / Lead Research Engineer

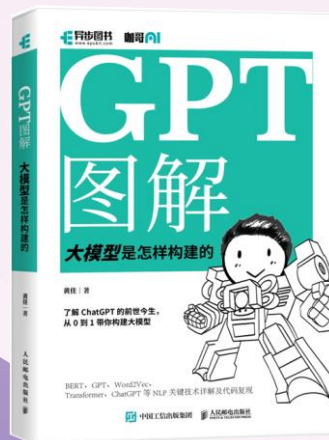
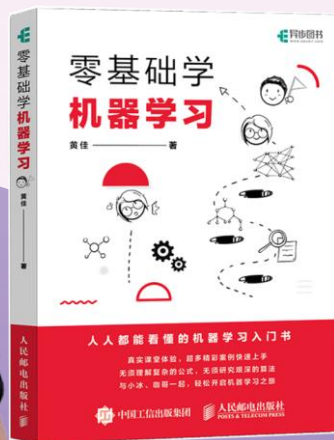
AiCon
全球人工智能开发与应用大会

黄佳

研究员 / 技术图书作者 / 极客时间专栏作者

《极客时间 LangChain实战课》

《极客时间 RAG进阶训练营》



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- 01 为什么我们仍然在谈论RAG
- 02 RAG落地痛点及优化思路
- 03 企业文档合规性问答系统落地实践
- 04 医疗术语标准化系统的落地实践
- 05 知识图谱在医疗术语标准化系统中的应用
- 06 MCP和A2A时代的RAG

📍 北京

QCon

全球软件开发大会

会议时间：4月10-12日

- 大模型赋能 AIOps
- 越挫越勇的大前端
- 云上业务架构演进
- 多模态大模型及应用
- 海外 AI 应用创新实践

📍 北京

AiCon

全球人工智能开发与应用大会

会议时间：6月27-28日

- 端侧智能
- AI Agent
- 多模态大模型
- 金融+大模型

📍 上海

QCon

全球软件开发大会

会议时间：10月23-25日

- AI Agent
- 大模型训练推理
- 端侧 AI
- 搜推深度融合
- 数智金融

4月

5月

6月

8月

10月

12月

📍 上海

AiCon

全球人工智能开发与应用大会

会议时间：5月23-24日

- 通用大模型
- AI Agent
- 垂直领域应用
- 模型可解释性

📍 深圳

AiCon

全球人工智能开发与应用大会

会议时间：8月22-23日

- 模型效率与部署
- 多模态大模型
- 大模型安全
- 智能硬件

📍 北京

AiCon

全球人工智能开发与应用大会

会议时间：12月19-20日

- 通用大模型
- 智能硬件
- LMOPs
- 具身智能

极客邦科技 2025 年会议规划

促进软件开发及相关领域知识与创新的传播



参会咨询



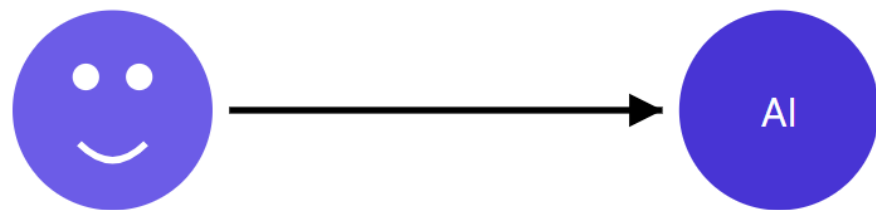
查看会议

01 为什么我们仍然谈论RAG

RAG的落地，
往往是面向业务做RAG，
而不是反过来面向RAG做业务。

—— 刘焕勇老师

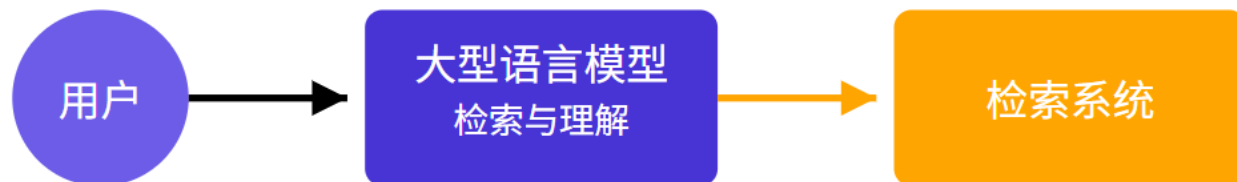
1. 人类与大模型直接对话



用户: "销售数据分析"

AI: "第三季度销售增长7.15%..."

2. 大模型进行知识检索



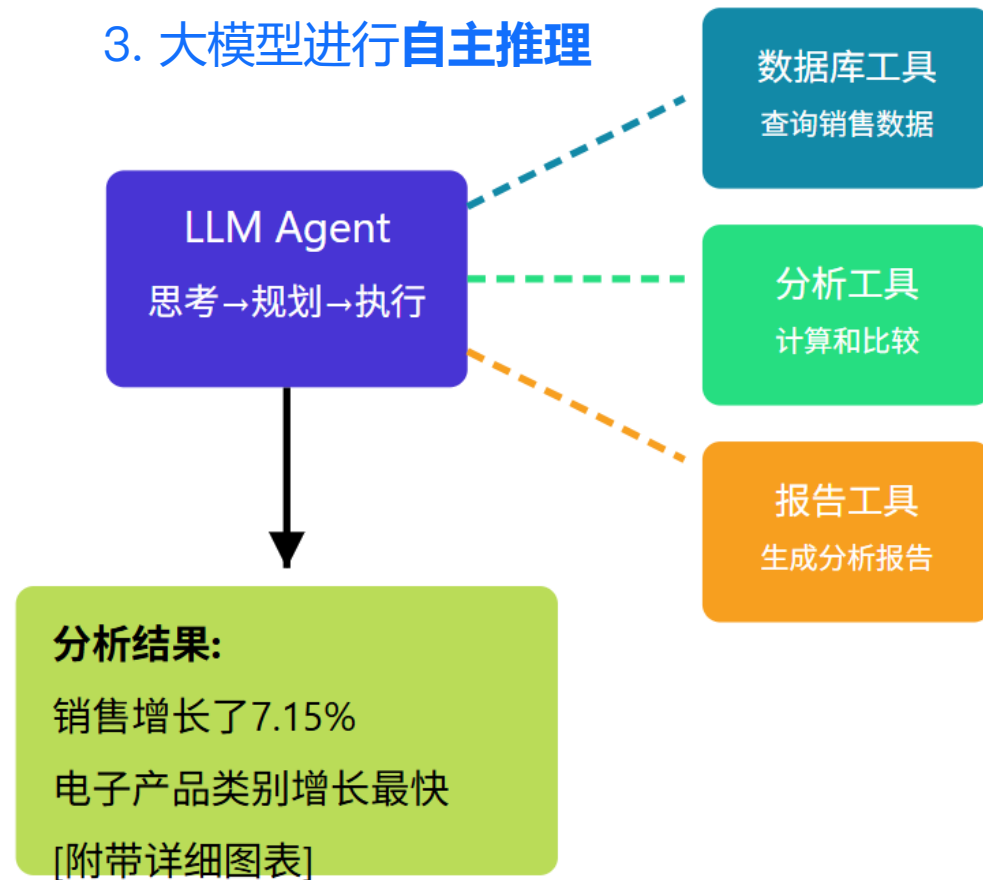
检索结果:

销售额: ¥1,250万

增长率: 17.3%

大模型时代的应用开发

3. 大模型进行自主推理



■ 大模型价值利用的两个通用范式

智能体的推理能力

Agent + Tool Calls

大模型时代的知识检索

RAG + Fine Tuning

■ RAG是“模型×知识×数据”融合的关键



■ RAG 看似简单，价值却难兑现



02 RAG落地痛点及优化思路

RAG落地难点

文档的导入和解析（图、表）

如何将相关联的内容整体切片或建立起相关联的索引

如何处理大规模、分布式向量数据

Text2SQL的精细化设计

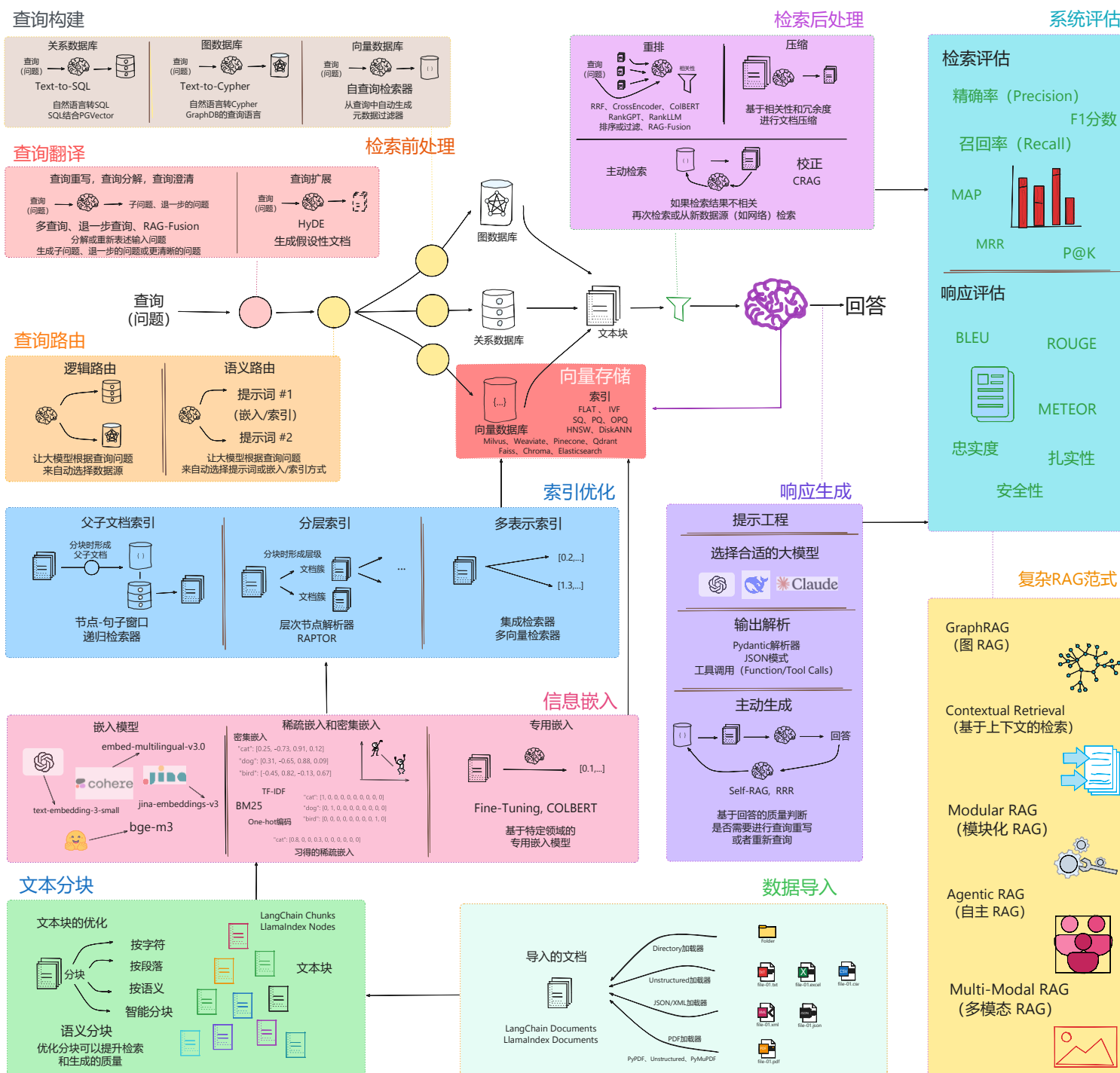
如何构建程序代码的检索系统

图数据库和知识图谱和RAG系统的结合

如何设计有权限的RAG系统

RAG系统优化思路

寻找瓶颈点



03 文档合规性问答系统落地实践

项目背景：可持续报告合规性分析

可持续发展报告（Sustainability Report）是企业或组织自愿发布的一种非财务报告，主要披露企业在环境、社会责任和治理（ESG）方面的绩效，并展现企业如何管理可持续发展风险和机遇。

1. 环境（E）

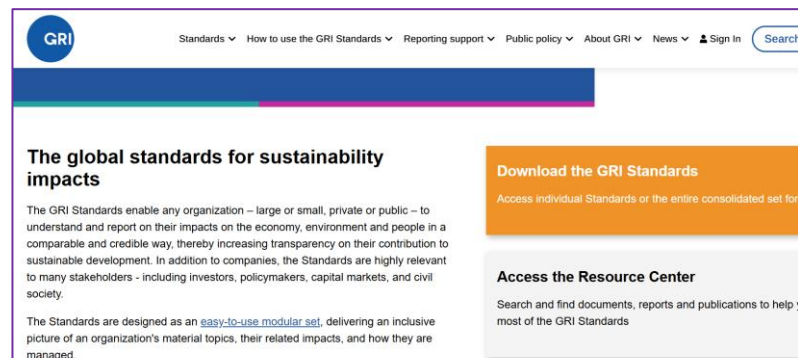
- 1.碳排放（Scope 1, Scope 2, Scope 3）
- 2.能源消耗、用水情况、废弃物管理
- 3.绿色产品和供应链管理

2. 社会（S）

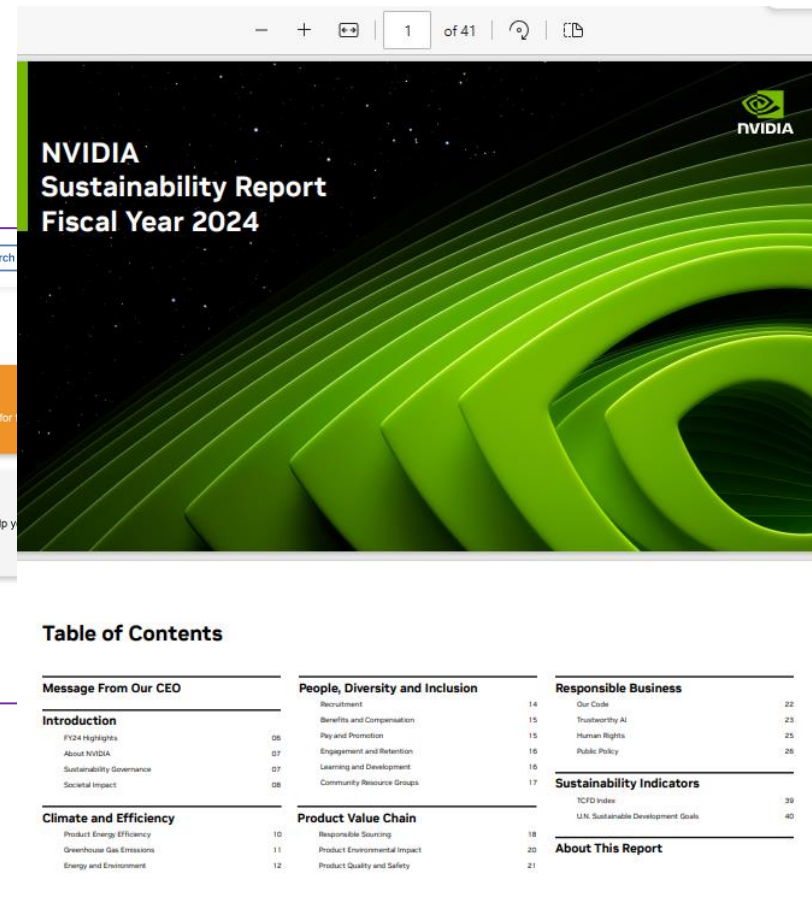
- 1.员工薪酬、健康和安全
- 2.多样性和包容性（D&I）
- 3.社区发展、客户隐私、数据保护

3. 治理（G）

- 1.反腐败政策、商业道德
- 2.董事会治理、股东权利
- 3.供应链合规



	2024-Telstra-Sustainability-Report.pdf	✓
	BHP_Sustainability_2024.pdf	✓
	FY2024-NVIDIA-Corporate-Sustainability-Repor...	✓
	microsoft_sustainability_report.pdf	✓
	USA Engineering Sustainability Report.pdf	✓



内部审计：企业发布的Sustainability Report究竟是否符合GRI标准

三个难点 + 痛点



- PDF格式不统一（文字、图片、公式、表格）
- 文档结构复杂（章节、标题、子标题、段落之间存在多层嵌套关系）
- 表格数据和正文密切相关，强行切分会破坏语义关联信息
- 分散在多种来源（网页、内部文档库、第三方数据库等）

文档导入



- 如何构建含层次关系的索引（2024年的数据要与其标题有隶属关系）
- 如何在构建索引时保持图表的原始结构化信息
- 如何在索引过程中管理文档的元数据信息，进行检索结果的精准过滤与排序

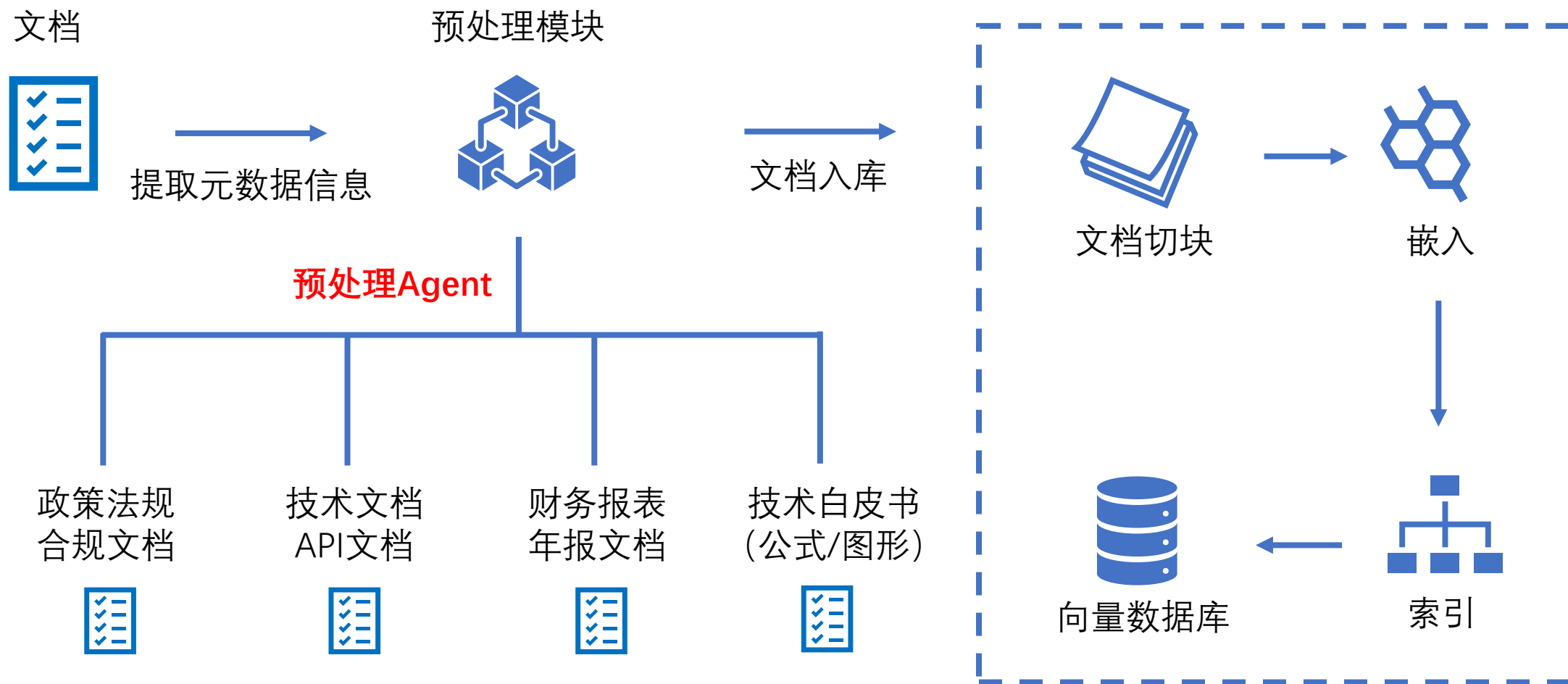
索引设计



- 缺乏统一、有效的评估指标体系
- 缺少高质量的评估数据集，导致评估结果不真实
- 难以追踪评估与改进的闭环，系统难以持续优化

如何评估

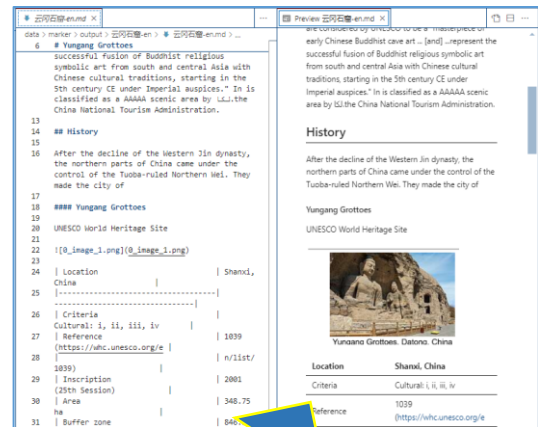
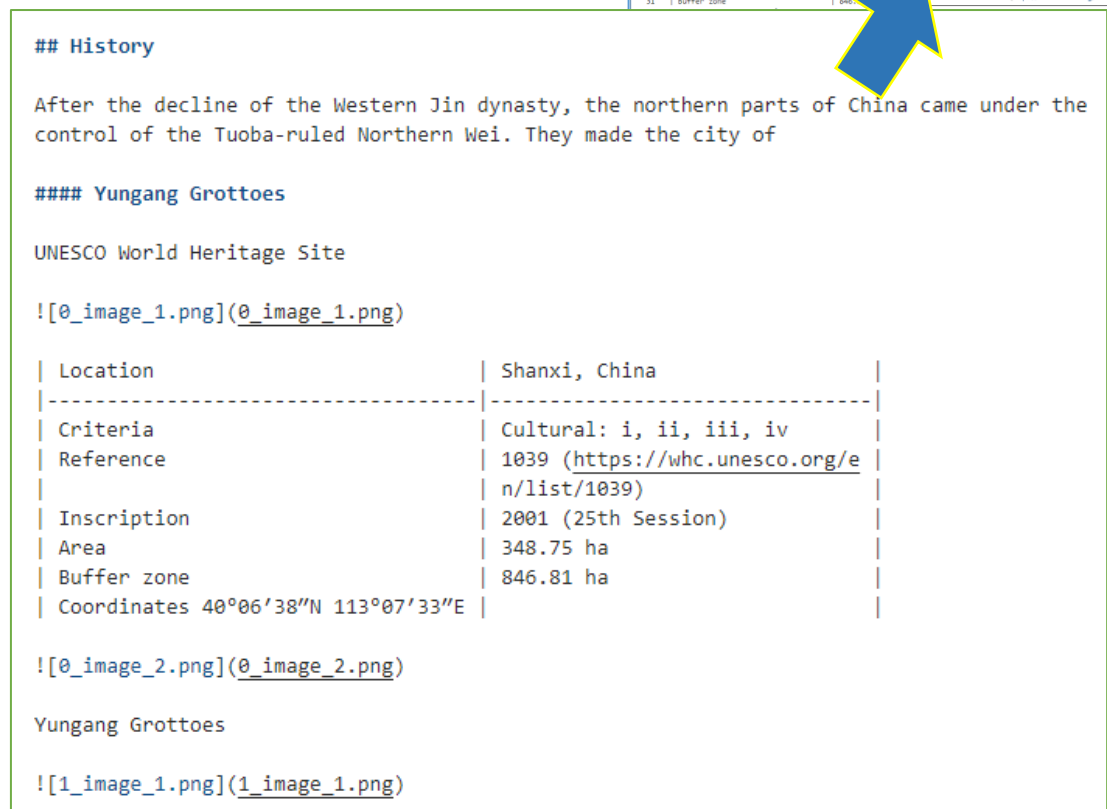
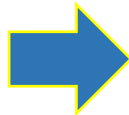
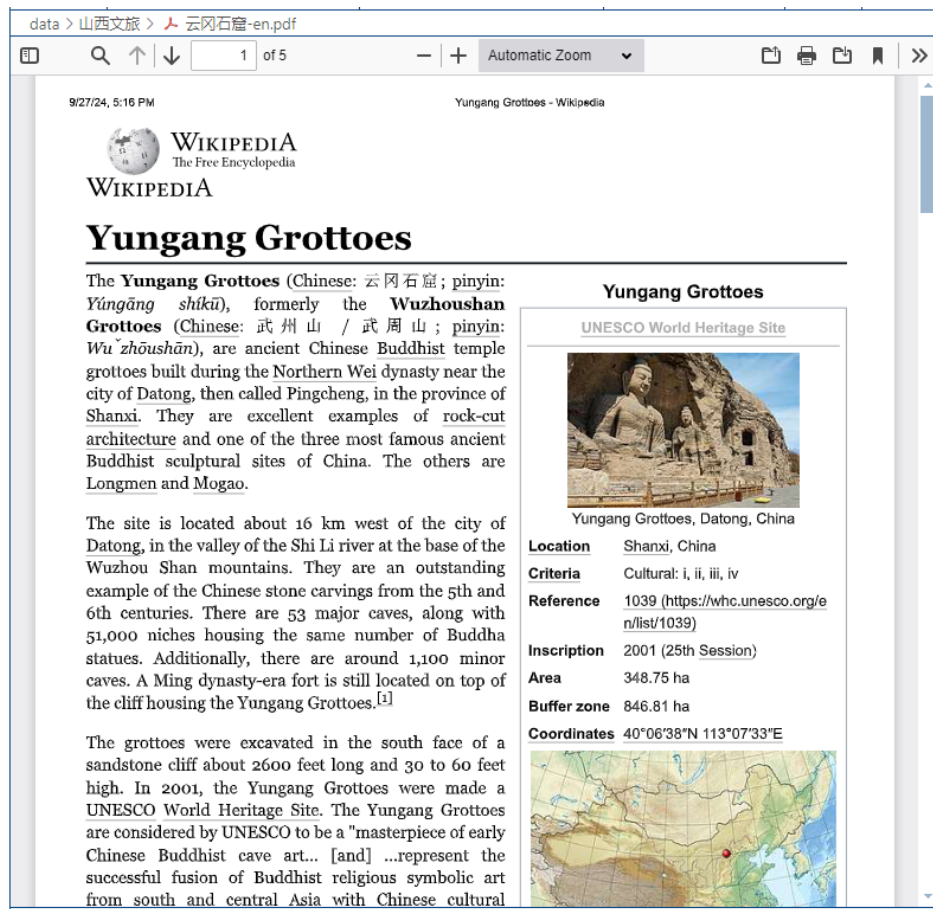
■ 难点1 文档的导入（知识库的构建）



■ 各种工具的简单比较

文档加载器	说明	Package/API	特点
PyPDF	使用 pypdf 加载和解析PDF文件	Package	高效轻量, 适合处理简单PDF文档
Unstructured	使用 Unstructured 的开源库加载PDF文件	Package/API	兼容多种文档格式, 支持内容提取和分析
Amazon Textract	使用 AWS API 加载PDF文件	API	云服务支持, 适合大批量文档的OCR处理
MathPix	使用 MathPix 加载和解析PDF文件	API	专为数学公式设计, 精准解析复杂内容
PDFPlumber	使用 PDFPlumber 加载PDF文件	Package	丰富的PDF内容控制和处理功能
PyPDFDirectry	加载目录中的PDF文件	Package	批量加载, 便于处理多个PDF文档
PyPDFium2	使用 PyPDFium2 加载PDF文件	Package	高效解析, 支持PDF页面的渲染和转换
PyMuPDF	使用 PyMuPDF 加载PDF文件	Package	速度优化, 支持复杂PDF的精细化处理
PDFMiner	使用 PDFMiner 加载PDF文件	Package	适合文本抽取, 处理PDF中的嵌入文字内容

多数情况需要把PDF转Markdown



重点是解析出元素之间的关系 — 父子元素的链接

```
from langchain_unstructured import UnstructuredLoader
from typing import List
from langchain_core.documents import Document
page_url = "https://zh.wikipedia.org/wiki/黑神话：悟空"
def _get_setup_docs_from_url(url: str) -> List[Document]:
    loader = UnstructuredLoader(web_url=url)
    setup_docs = []
    # parent_id = None # 初始化 parent_id
    # current_parent = None # 用于存储当前父元素
    for doc in loader.load():
        # 检查是否是 Title 或 Table
        if doc.metadata["category"] == "Title" or
doc.metadata["category"] == "Table":
            parent_id = doc.metadata["element_id"]
            current_parent = doc # 更新当前父元素
            setup_docs.append(doc)
        elif doc.metadata.get("parent_id") == parent_id:
            setup_docs.append((current_parent, doc)) # 将父元素和子
元素一起存储
    return setup_docs
```

除Markdown之外，我还需要构建一套索引系统

Title: 参考资料

父元素 - Title: 参考资料
子元素 - UncategorizedText: [编辑]

父元素 - Title: 参考资料
子元素 - ListItem: ^ GNN 記者 Edward. 「西遊記」改編動作 RPG《黑神話：悟空》釋出
(臺灣)) .

父元素 - Title: 参考资料
子元素 - ListItem: ^ 如今我风华正茂：游戏科学是如何制造《黑神话：悟空》的 (页1

父元素 - Title: 参考资料
子元素 - ListItem: ^ Chinese action game Black Myth: Wukong is taking the inter

父元素 - Title: 参考资料
子元素 - ListItem: ^ 解园, 姜溟然. 中国首款3A游戏发布再掀传统文化热潮. 新华社.

和表头关联非常密切的PDF表

这个PDF是一个 带表格的结构化数据，
所以直接用SimpleDirectoryReader加载
并用SentenceSplitter进行分块后，信息
可能被破坏，从而导致问答效果不佳。

2021

In the 35th annual *Forbes* list of the world's billionaires, the list included 2,755 billionaires with a total net wealth of \$13.1 trillion, up 660 members from 2020; 86% of these billionaires had more wealth than they possessed last year.^{[11][12]}

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5	Mark Zuckerberg	\$97 billion	36	United States	Meta Platforms
6	Warren Buffett	\$96 billion	90	United States	Berkshire Hathaway
7	Larry Ellison	\$93 billion	76	United States	Oracle Corporation
8	Larry Page	\$91.5 billion	48	United States	Alphabet Inc.
9	Sergey Brin	\$89 billion	47	United States	Alphabet Inc.
10	Mukesh Ambani	\$84.5 billion	63	India	Reliance Industries

2020

https://en.wikipedia.org/wiki/The_World%27s_Billionaires

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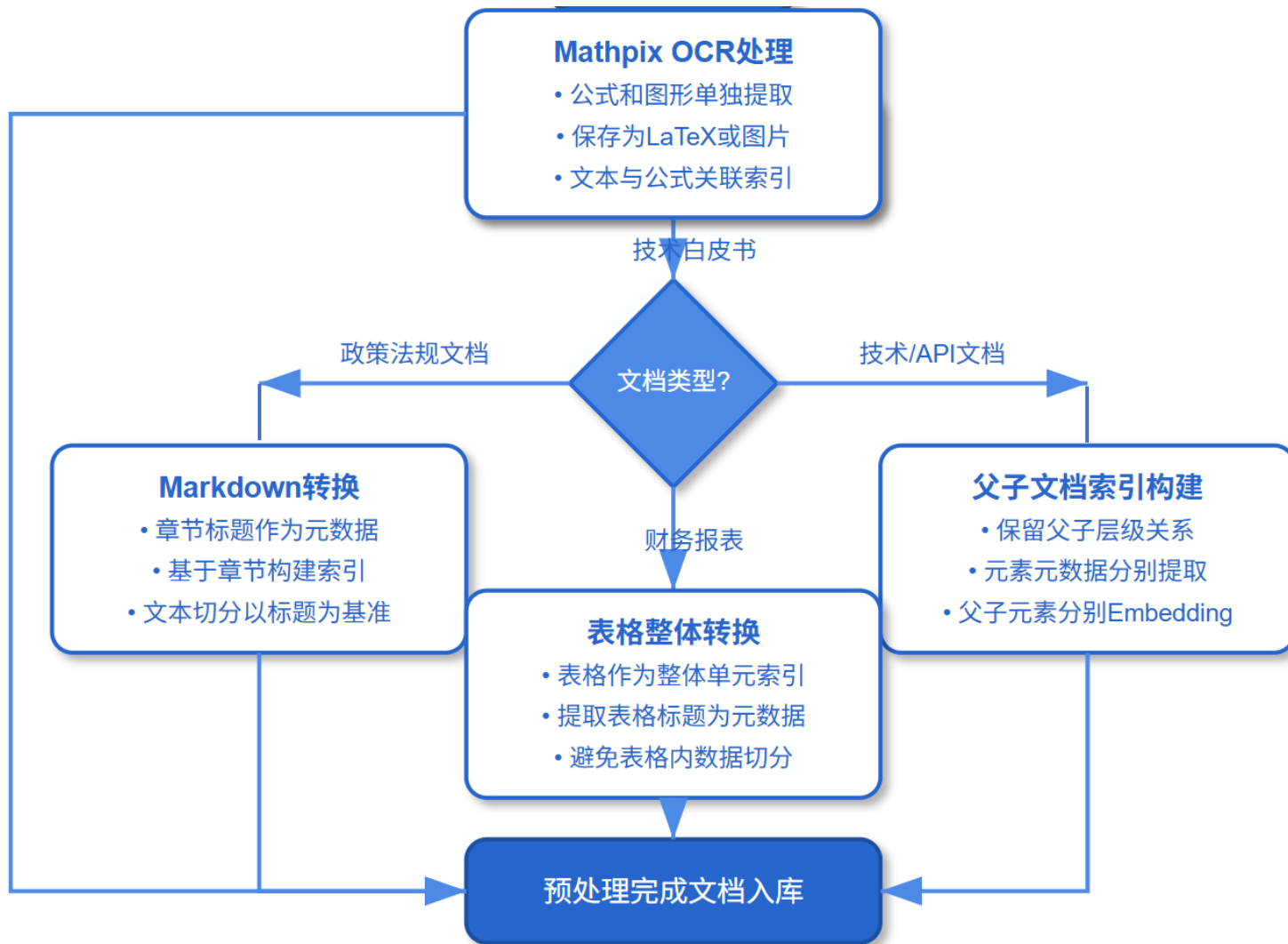
The World's Billionaires - Wikipedia

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5	Larry Ellison	\$59 billion	75	United States	Oracle Corporation
6	Amancio Ortega	\$55.1 billion	84	Spain	Inditex , Zara
7	Mark Zuckerberg	\$54.7 billion	35	United States	Facebook, Inc.
8	Jim Walton	\$54.6 billion	71	United States	Walmart
9	Alice Walton	\$54.4 billion	70	United States	Walmart
10	S. Robson Walton	\$54.1 billion	77	United States	Walmart

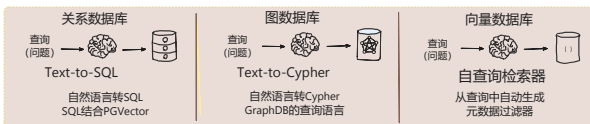
2019

■ 文档预处理Agent决策流程

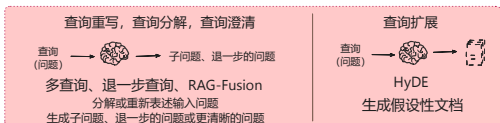


寻找瓶颈点

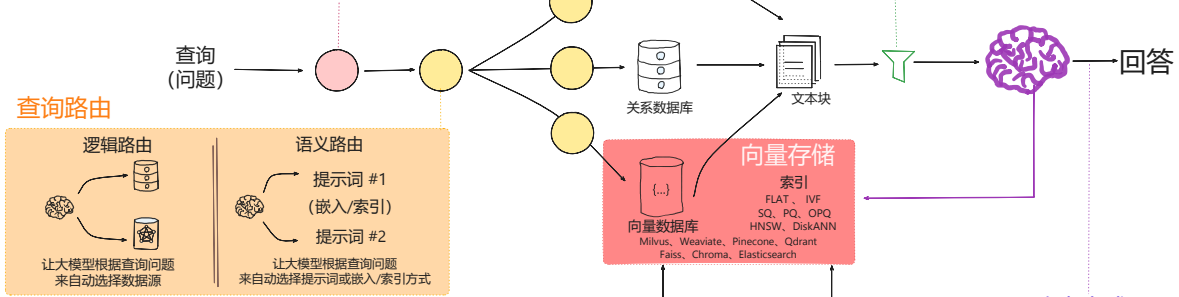
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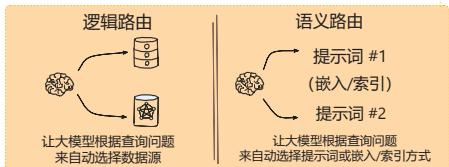
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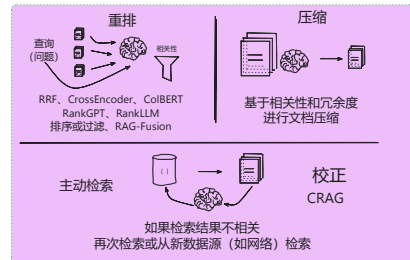
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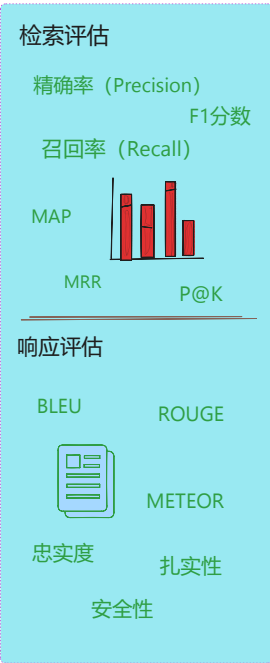
查询路由



检索后处理



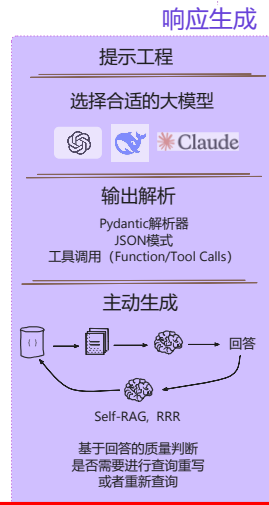
系统评估



索引优化

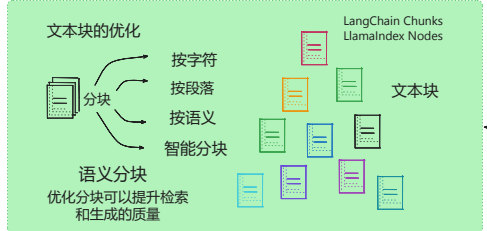


信息嵌入

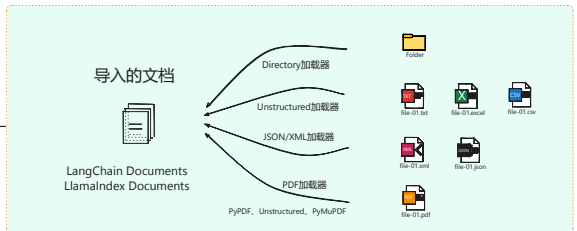


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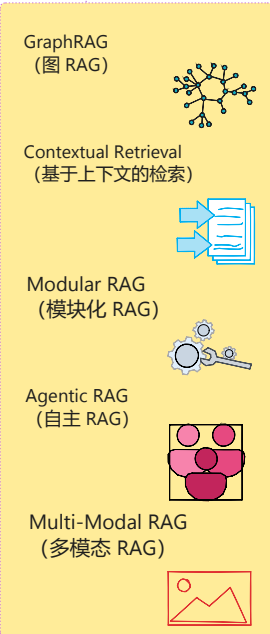
文本分块



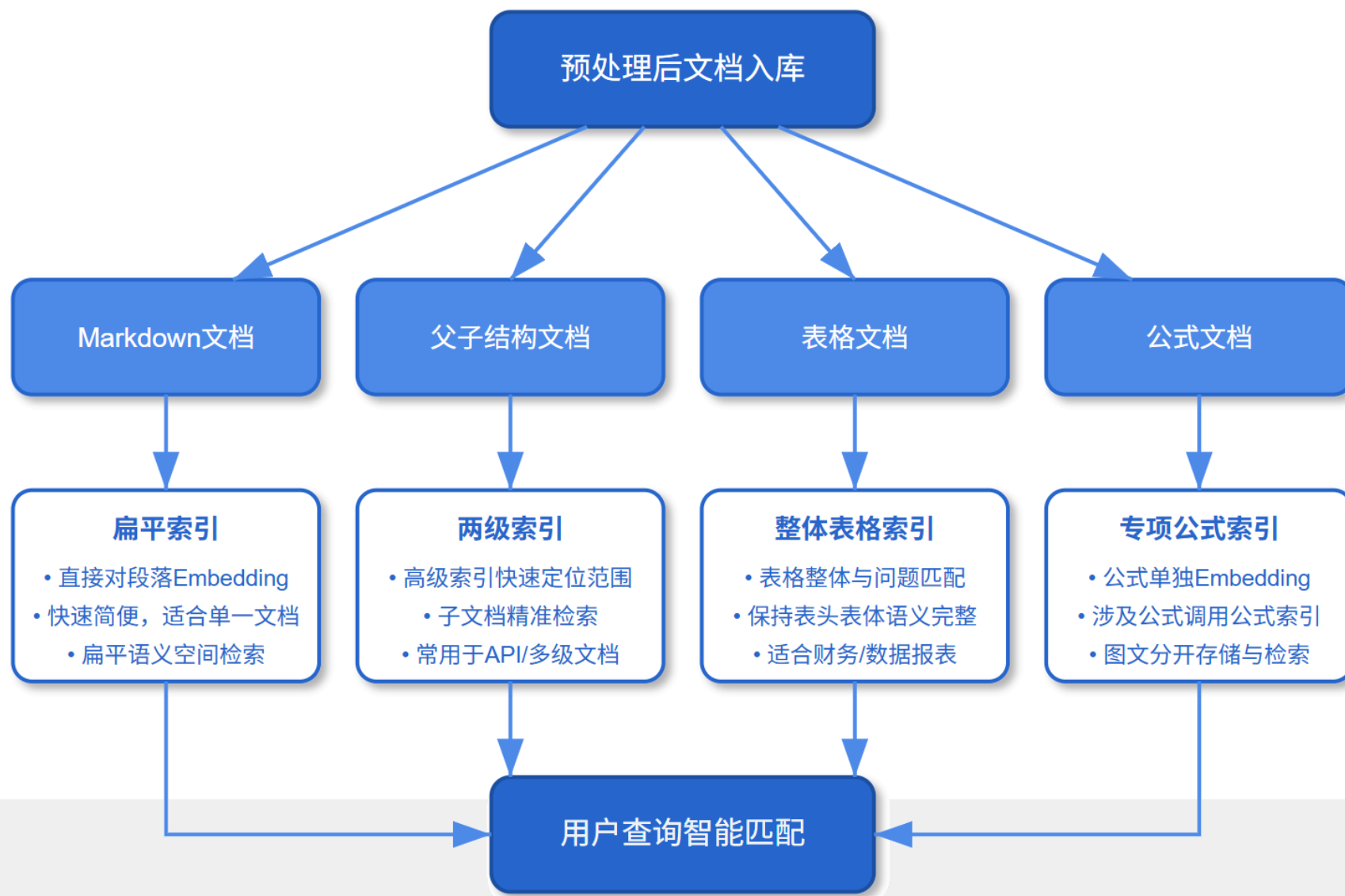
数据导入



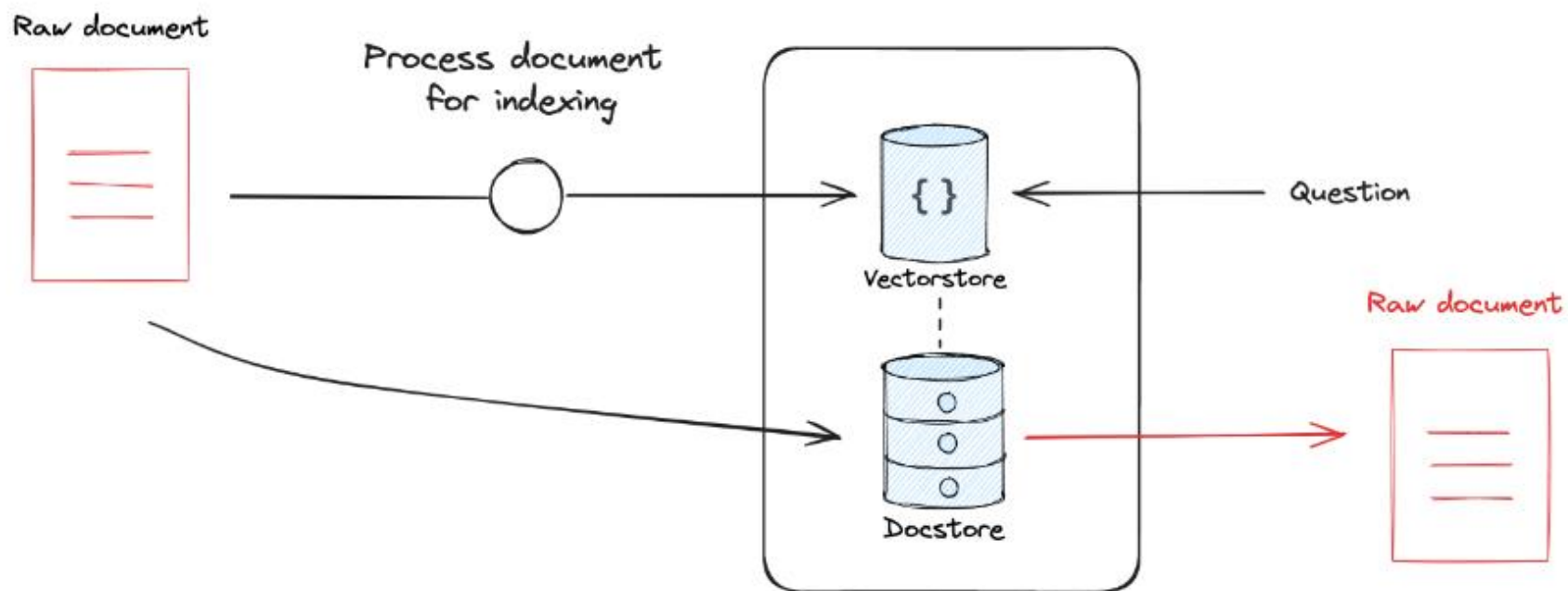
复杂RAG范式



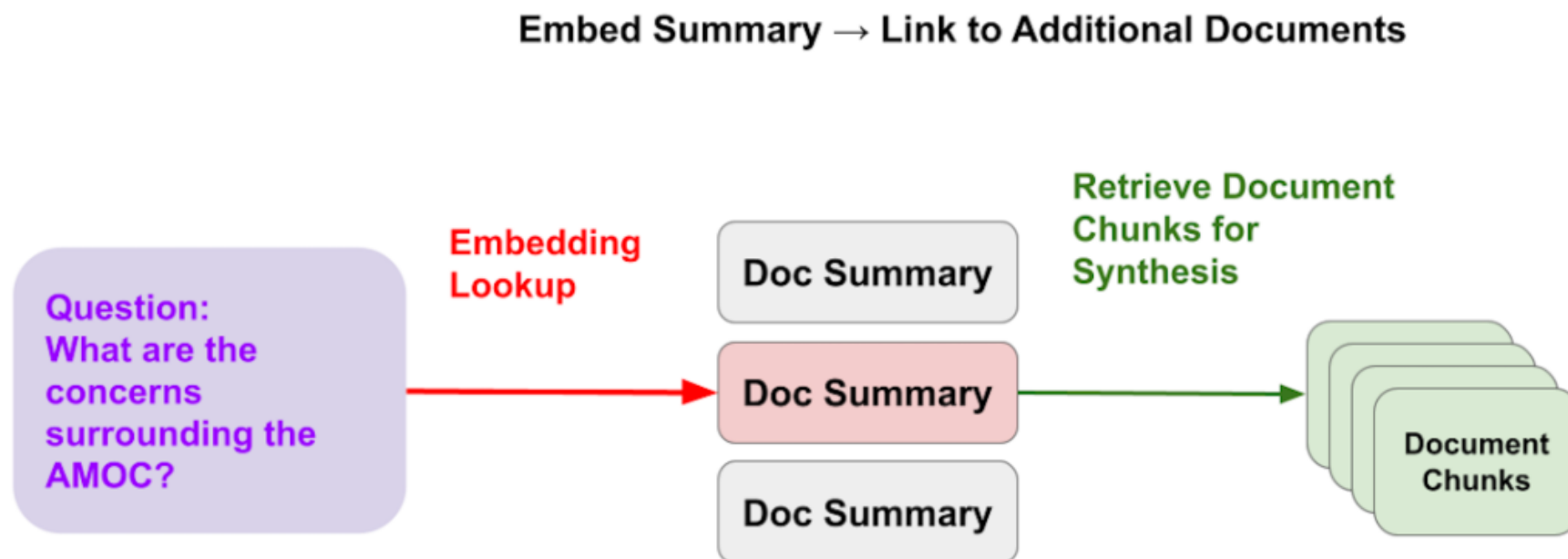
■ 难点2 索引优化（层次化索引的构建）



父子文本块检索



■ 构建有层次的索引



如何实现：多种方法

1. 构建两个向量数据库（Summary 和 Details），通过 Metadata 进行链接
2. 通过 LlamaIndex 的 IndexNode 和 PandasQueryEngine
3. 也可以通过查询先检索相关表名，然后做 Text2SQL
4. 对于这个例子，也可以提取年份，用元数据进行 Filter

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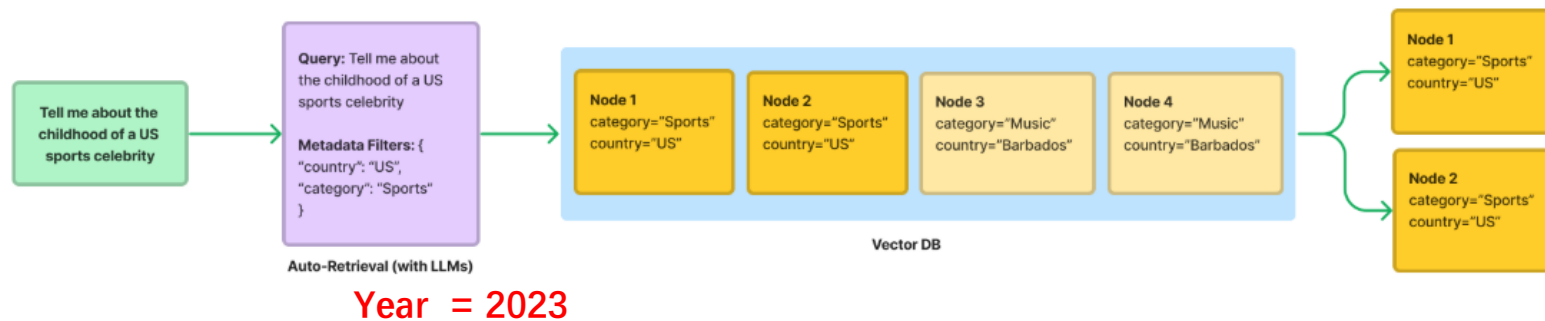
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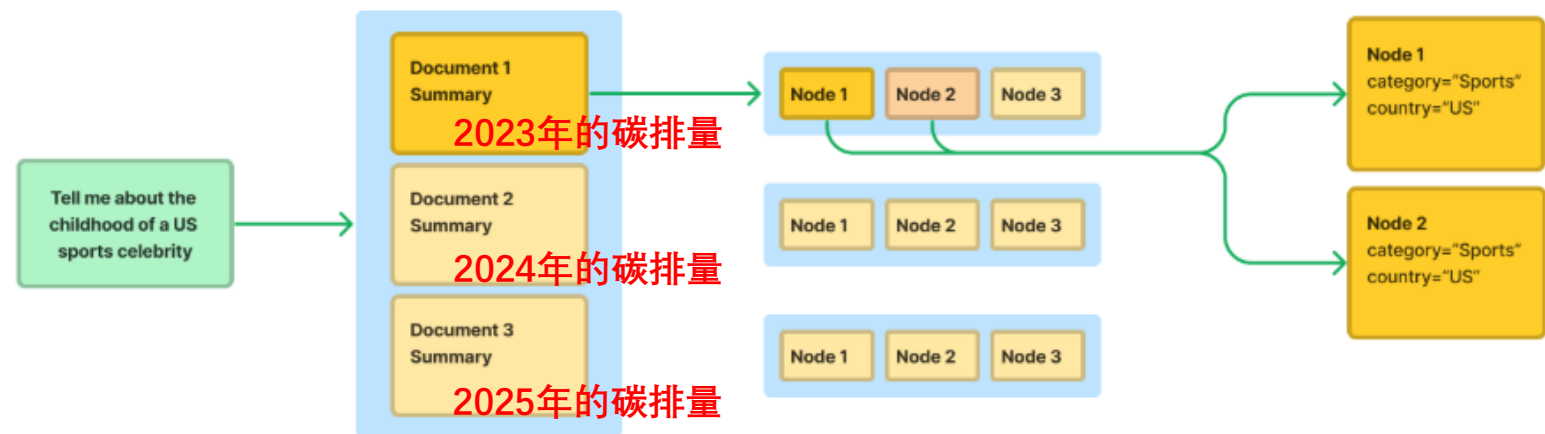
2019



思路1: 元数据提取

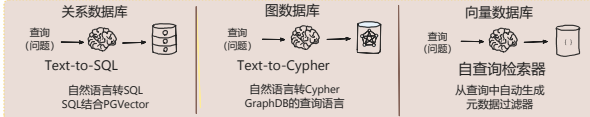


思路2: 直接检索Summary节点

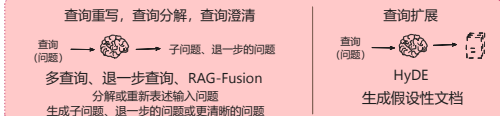


寻找瓶颈点

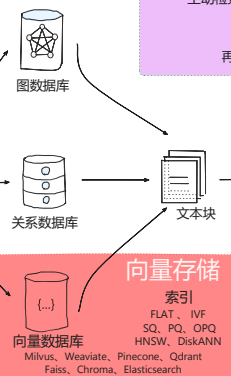
查询构建



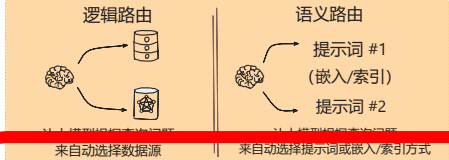
查询翻译



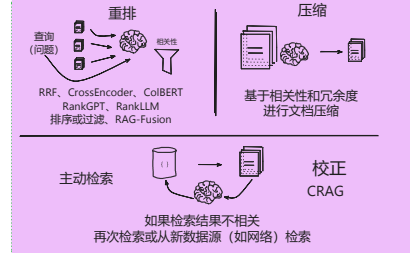
检索前处理



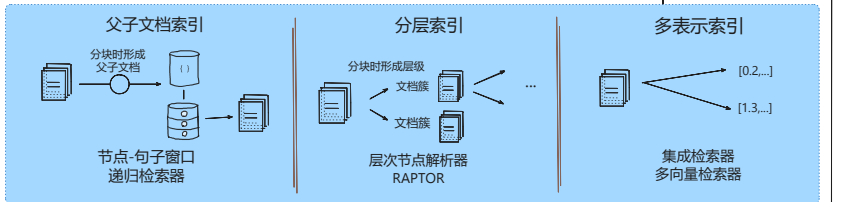
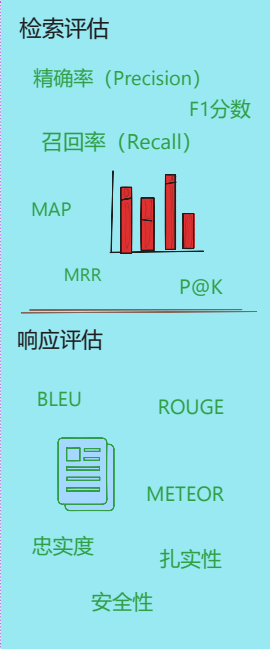
查询路由



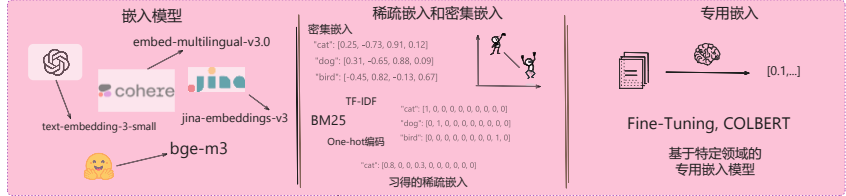
检索后处理



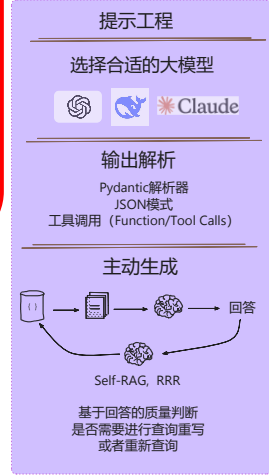
系统评估



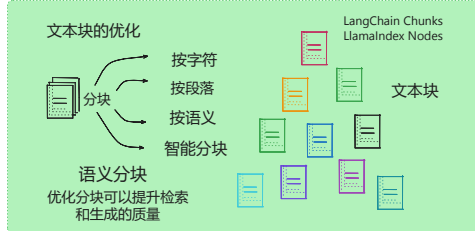
信息嵌入



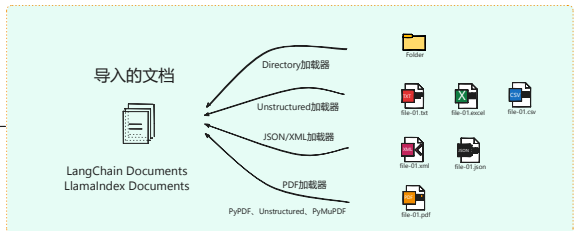
响应生成



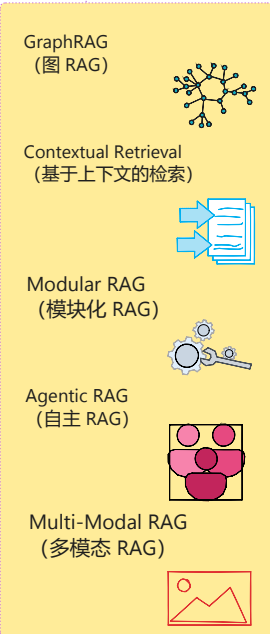
文本分块



数据导入



复杂RAG范式



■ 难点3 – 如何评估合规系统

- 评估数据集应该如何构建
- 如何设计有针对性的评估机制

Message From Our CEO Introduction Climate and Efficiency People, Diversity and Inclusion Product Value Chain Responsible Business Sustainability Indicators		
People, Diversity and Inclusion		
Metric	FY24	Reference Indicator
Employee Health and Safety		
Occupational health and safety	Our Health and Safety (H&S) team oversees workplace conditions for NVIDIANS globally. H&S team members provide guidance to ensure that facilities meet or exceed local safety requirements, promote safe work practices, and support compliance with applicable health and safety legislation and policies. Our H&S framework includes: > A network of office-level committees and site safety officers. > Globally applicable programs on topics including ladder safety, ergonomics, vehicle safety, and chemical management. > Role-specific online training courses such as hazardous waste, data center safety, lab safety, and site safety officer training. We identify and assess hazards and risks through routine inspections, external audits, incident reporting, and investigations. Incidents and the results of incident investigations are recorded in our global security database. Our processes for hazard assessment and incident investigation are documented in our Illness and Injury Prevention Program.	GRI 403-1, 403-4, 403-5, 403-7, 403-8 UNGC L1, L2, L3, L4, L5 SASB TC-SC-320a.1
Total amount of monetary losses as a result of legal proceedings associated with employee health and safety violations	We disclose information on legal proceedings in our quarterly reports on Form 10-Q and our Annual Reports on Form 10-K. In FY24, there were no instances associated with employee health and safety violations.	SASB TC-SC-320a.2

检索评估

精确率Precision

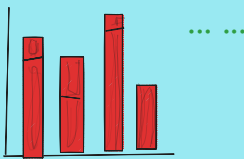
F1分数

召回率Recall

MAP

MRR

P@K



生成评估

BLEU

ROUGE



METEOR

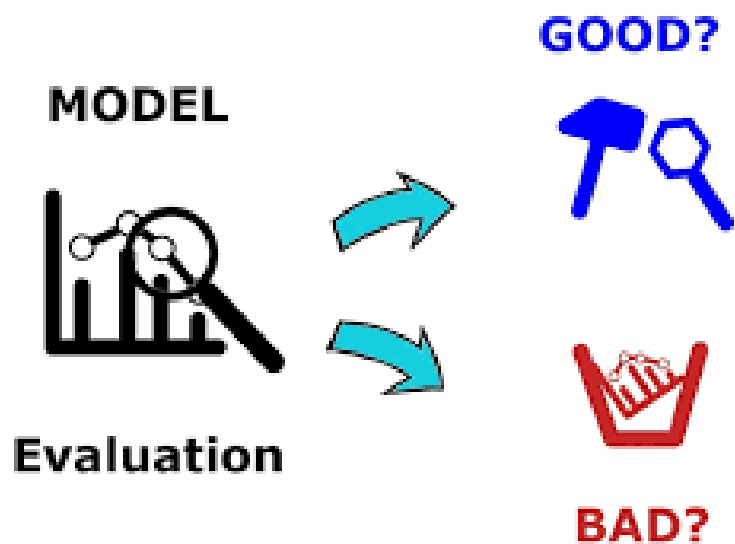
忠实度

扎实性

安全性



任何系统都需要有评估机制



机器学习性能评估指标

分类

Accuracy

Precision

Recall

F1 Score

ROC Curve

PR Curve

AUC

回归

MAE

MSE

聚类

ARI --- Adjusted Rand index

轮廓系数 Silhouette Coefficient

有一系列客观评估工具



Supercharge Your LLM Application Evaluations

release v0.2.15 Made with Python license Apache-2.0 downloads 408k/month chat 161 online

Documentation | Quick start | Join Discord | Blog | Newsletter | Careers

trulens v1.4.0 Azure Pipelines succeeded license MIT downloads 52k/month discourse 445k users docs trulens.org Open in Colab

Welcome to TruLens!



id	name	input	output	evaluation	start
1	query	What is the impact of climate change on the environment?	The impact of climate change on the environment is significant. It leads to rising sea levels, more frequent and severe weather events, and the loss of biodiversity.	4.0/5	17/03/2024, 17:18
2	query	What is the significance of the Paris Agreement?	The Paris Agreement is a landmark treaty that aims to limit global warming to well below 2°C, preferably to 1.5°C, compared to pre-industrial levels.	4.0/5	17/03/2024, 17:18
3	query	How do you use TruLens?	TruLens is used to evaluate the performance of LLM applications. It provides a suite of metrics to measure the quality of the model's outputs.	4.0/5	17/03/2024, 17:18
4	query	What is the purpose of the Paris Agreement?	The purpose of the Paris Agreement is to keep the global average temperature rise this century well below 2°C, preferably to 1.5°C, compared to pre-industrial levels.	4.0/5	17/03/2024, 17:18
5	embedding	-	-	4.0/5	17/03/2024, 17:18
6	synthetics	What is the purpose of the Paris Agreement?	The purpose of the Paris Agreement is to keep the global average temperature rise this century well below 2°C, preferably to 1.5°C, compared to pre-industrial levels.	4.0/5	17/03/2024, 17:18
7	chatting	-	-	4.0/5	17/03/2024, 17:18
8	chatting	-	-	4.0/5	17/03/2024, 17:18
9	chatting	-	-	4.0/5	17/03/2024, 17:18
10	chatting	-	-	4.0/5	17/03/2024, 17:18

DeepEval.

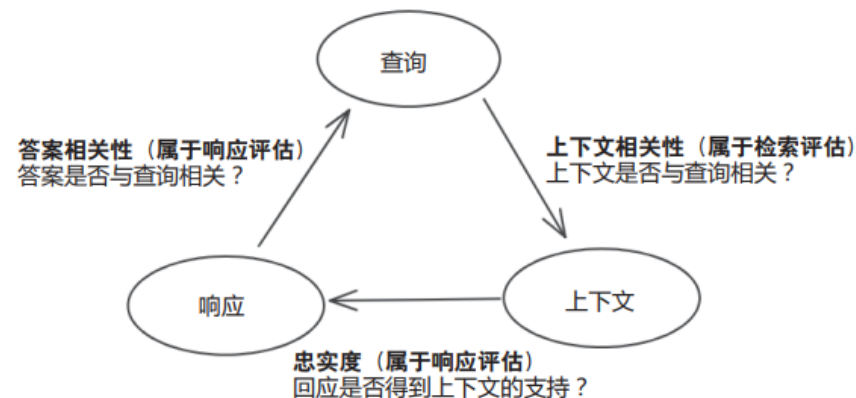
The LLM Evaluation Framework

DeepEval (by Confidant AI) 2161 members

Documentation | Metrics and Features | Getting Started | Integrations | DeepEval Platform

release v0.8.0 Open in Colab license Apache-2.0

DeepEval is a simple-to-use, open-source LLM evaluation framework, for evaluating and testing large-language model systems. It is similar to Pytest but specialized for unit testing LLM outputs. DeepEval incorporates the latest research to evaluate LLM outputs based on metrics such as G-Eval, hallucination, answer relevancy, RAGAS, etc., which uses LLMs and various other NLP models that runs locally on your machine for evaluation.



真实性 (幻觉)
忠实度 (上下文)
扎实性 (相关性)
安全性 (对齐)

整体来说，
这些工具用处不大

■ 重要的是构建专属的评估数据集和评估规则

	A	B	C	D	E
1	ID	Disclosure Requirement	Corresponding Text	Page Number	Compliance Status
2	301-1	Materials used by weight or volume	In FY24, over 90% of our NVIDIA GPU systems packaging was recyclable n	20	Y
3	301-2	Recycled input materials used	We embrace opportunities to reduce packaging materials and increase t	20	Y
4	305-1	Direct (Scope 1) GHG emissions	Scope 1 and 2: By the end of FY25, and annually thereafter, we expect to a	11,12,20,29,30,39,40	Y
5	305-2	Energy indirect (Scope 2) GHG emissions	To manage the GHG emissions footprint of our data centers, labs, and of	11,12,20,29,30,39,40	Y
6	305-3	Other indirect (Scope 3) GHG emissions	By the end of FY26, we expect to engage manufacturing suppliers compris	11,12,20,29,30,39,40	Y
7	306-1	Waste generation and significant waste-related impacts	We aim to reduce the amount of waste we send to landfill through waste i	12	Y
8	306-3	Significant spills			N
9	302-4	Reduction of energy consumption	We increased the amount of our renewable electricity use to 76%, through	12	Y
10	303-3	Water withdrawal	Water withdrawal: 382,636 m ³ in FY24, monitored under sustainability ir	32	Y
			Our Health and Safety (H&S) team oversees workplace conditions for NVIDIA's globally. H&S team		
11	403-1	Occupational health and safety management system	members provide guidance to ensure that facilities meet or exceed local	34	Y

评估规则因项目和任务而异

检索评估

- 精确率
- 召回率
- F1分数
- 平均倒数排名
- 平均精确率
- P@K

生成评估

- BLEU
- ROUGE
- METEOR

评估

CSV文件

Choose File

NVIDIA_GRI...set_English.csv

集合

NVIDIA_openai_20250319235505 (41)

Top K 结果数

3

相似度阈值: 0.3

评估

统计摘要

平均命中分数

55.6%

已找到页面中正确页面的比例

平均查找分数

81.0%

已找到正确页面的比例

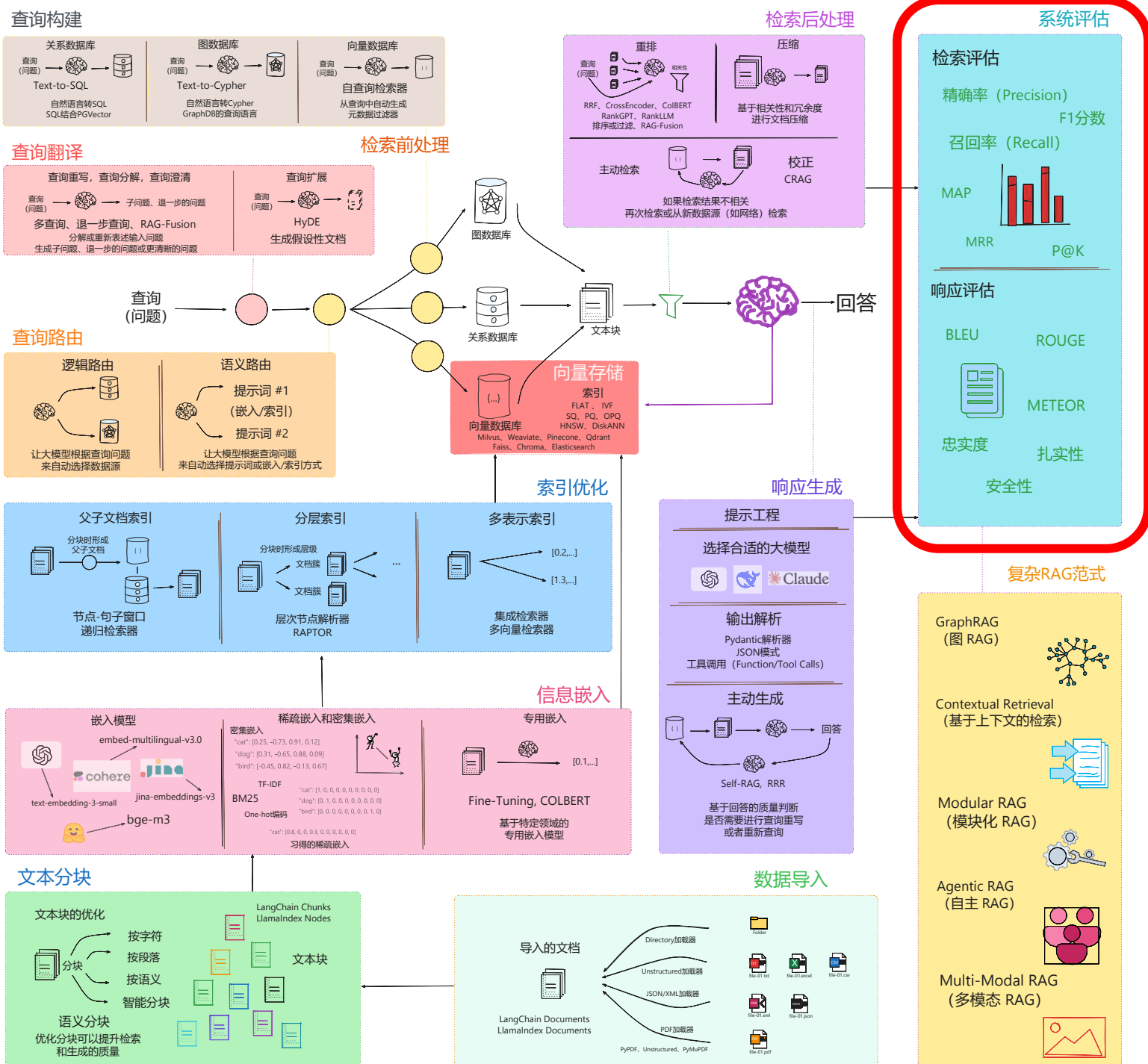
总查询数

9

详细结果

ID	要求	预期页码	找到页码	命中分数	查找分数	合规状态
301-1	Materials used by weight or volume	20	20, 12, 31	33.3%	100.0%	Y
301-2	Recycled input materials used	20	31, 20, 30	33.3%	100.0%	Y
305-1	Direct (Scope 1) GHG emissions	11, 12, 20, 29, 30, 39, 40	29, 11, 12	100.0%	42.9%	Y
305-2	Energy indirect (Scope 2) GHG emissions	11, 12, 20, 29, 30, 39, 40	29, 30, 12	100.0%	42.9%	Y
305-3	Other indirect (Scope 3) GHG emissions	11, 12, 20, 29, 30, 39, 40	29, 11, 20	100.0%	42.9%	Y
306-1	Waste generation and significant waste-related impacts	12	31, 12, 20	33.3%	100.0%	Y
302-4	Reduction of energy consumption	12	30, 11, 12	33.3%	100.0%	Y
303-3	Water withdrawal	32	32, 30, 31	33.3%	100.0%	Y
403-1	Occupational health and safety management system	34	34, 33, 37	33.3%	100.0%	Y

寻找瓶颈点



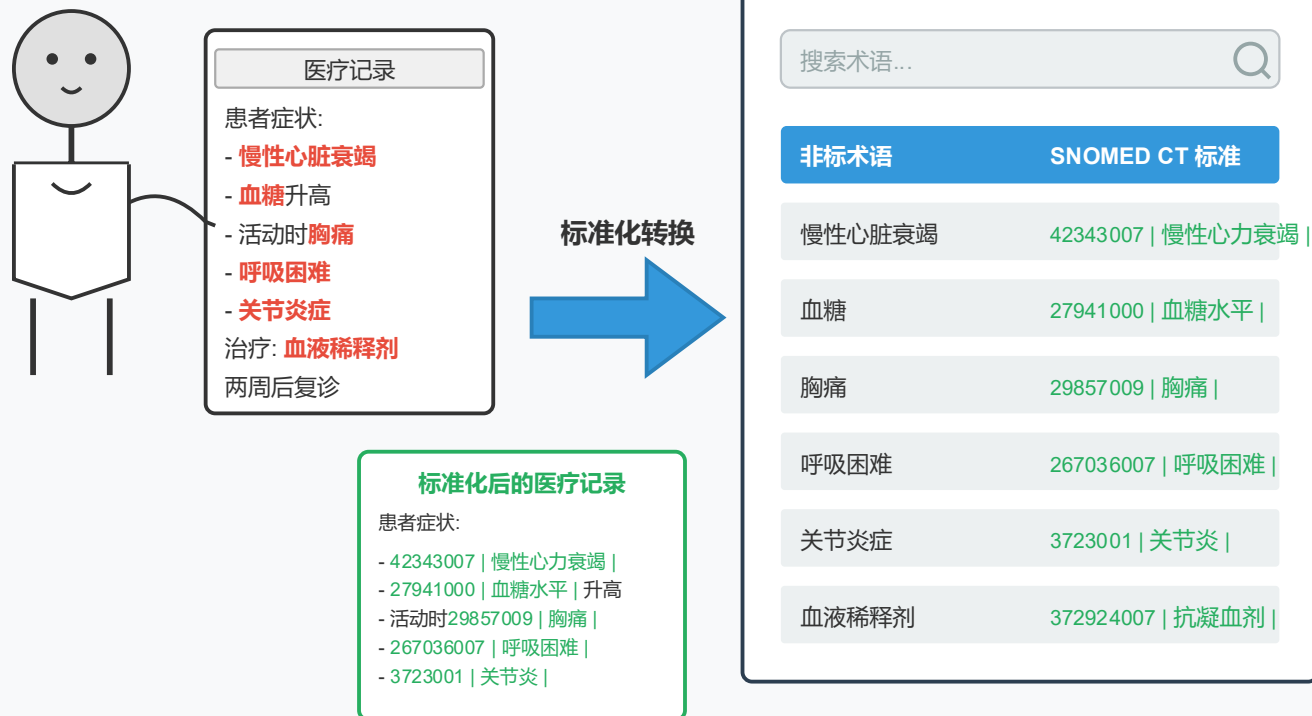
04 医疗术语标准化系统落地实践

一个看似简单的任务

医院内存在多种电子病历系统与数据标准
临床医生使用非标准化术语记录病情
医疗数据分析需要统一术语标准以提高准确性

核心挑战

- 专业术语多样性：同义词、缩写、俚语并存
- 领域知识壁垒：需要专业医学背景解读上下文
- 系统适应性：需应对不同科室、不同记录习惯
- 实时性要求：诊疗过程中需快速响应



- Is the **most comprehensive**, multilingual clinical healthcare terminology in the world
- Is a resource with comprehensive, scientifically **validated clinical content**
- Enables **consistent representation** of clinical content in electronic health records
- Is **mapped to other international standards**
- Is in use in **more than eighty countries**

The SNOMED CT clinical terminology has unmatched depth, enabling clinicians to record data with enhanced accuracy and consistency. SNOMED CT remains a growing and evolving product made better by the Community of Practice.

SNOMED CT supports the development of comprehensive high-quality clinical content in electronic health records. It provides a standardized way to represent clinical phrases captured by the clinician and enables automatic interpretation of these



挑战：如何连接看似语义无关的医学词汇

The screenshot displays the SNOMED CT Browser interface. The search bar at the top contains the text 'SOB', which is highlighted with a red box. Below the search bar, the results are listed in a table. The first result is 'Sobbing' (Crying (observable entity)). Other results include 'Sobrerol', 'Sobradhino', 'Soboliphyme', 'Aeromonas sobria', 'Sobemovirus group', 'Sobbing respiration', 'Sobrerol (substance)', 'Soboliphyme baturini', 'Sobrerol only product', 'Soboliphyme (organism)', 'Streptococcus sobrinus', 'Aeromonas sobria complex', 'SOB - Shortness of breath', 'Aeromonas sobria (organism)', 'Sobrerol-containing product', 'Sobemovirus group (organism)', and 'Sobradhino (qualifier value)'. The 'SOB - Shortness of breath' result is highlighted with a red box. On the right side of the interface, the 'Concept Details' panel is open, showing the details for 'Dyspnea' (finding), which is also highlighted with a red box. The panel includes a list of parents (Difficulty breathing (finding)), a list of children (39 Children), and a list of synonyms (Dyspnea, Dyspnoea, Breathless, Breathlessness, SOB - Shortness of breath, Shortness of breath, Dyspnea (finding)).

SNOMED CT Browser

Release: United Kingdom Edition v20250212

Perspective: Full

About

© SNOMED International 2017 v1.36.4 - Hosted and maintained by NHS Digital

Taxonomy Search Favorites Refset

Search

Options

Search Mode: Partial matching search mode

Status: Active components only

Group by concept

Filter results by Language

english 33

Filter results by Semantic Tag

organism 17

finding 7

medicinal product 4

substance 2

qualifier value 2

observable entity 1

Filter results by Module

SNOMED CT core module (core metadata concept) 33

Filter results by Refset

CTV3 to SNOMED CT simple map reference set (foundation metadata concept) 33

National Health Service Care Record Element association 14

Type at least 3 characters Example: shou fra

SOB

33 matches found in 0.203 seconds.

Concept	Description
Sobbing	Crying (observable entity)
Sobrerol	Sobrerol (substance)
Sobradhino	Sobradhino (qualifier value)
Soboliphyme	Soboliphyme (organism)
Aeromonas sobria	Aeromonas sobria (organism)
Sobemovirus group	Sobemovirus group (organism)
Sobbing respiration	Sobbing respiration (finding)
Sobrerol (substance)	Sobrerol (substance)
Soboliphyme baturini	Soboliphyme baturini (organism)
Sobrerol only product	Product containing only sobrerol (medicinal product)
Soboliphyme (organism)	Soboliphyme (organism)
Streptococcus sobrinus	Streptococcus sobrinus (organism)
Aeromonas sobria complex	Aeromonas sobria complex (organism)
SOB - Shortness of breath	Dyspnea (finding)
Aeromonas sobria (organism)	Aeromonas sobria (organism)
Sobrerol-containing product	Product containing sobrerol (medicinal product)
Sobemovirus group (organism)	Sobemovirus group (organism)
Sobradhino (qualifier value)	Sobradhino (qualifier value)

Concept Details

Summary Details Diagram Expression Refsets Members

References Classification Map

Stated Inferred

Parents

Difficulty breathing (finding)

Dyspnea

SCTID: 267036007

267036007 | Dyspnea (finding) |

Dyspnea

Dyspnoea

Breathless

Breathlessness

SOB - Shortness of breath

Shortness of breath

Dyspnea (finding)

Finding site → Structure of respiratory system

Interprets → Ease of respiration

Children (39)

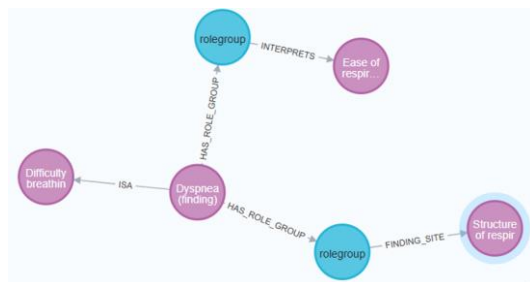
39 Children

05 知识图谱在医疗术语 标准化系统中的应用

构建并利用SNOMED图数据库



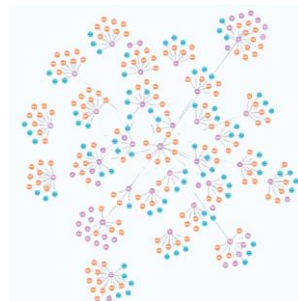
查找某医学概念的所有属性关系
(如“部位”、“过程”、“因果”)



因果关系



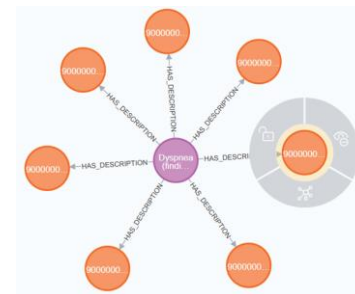
获取一个医学概念所有的上下文
(语义网络)



语义网络

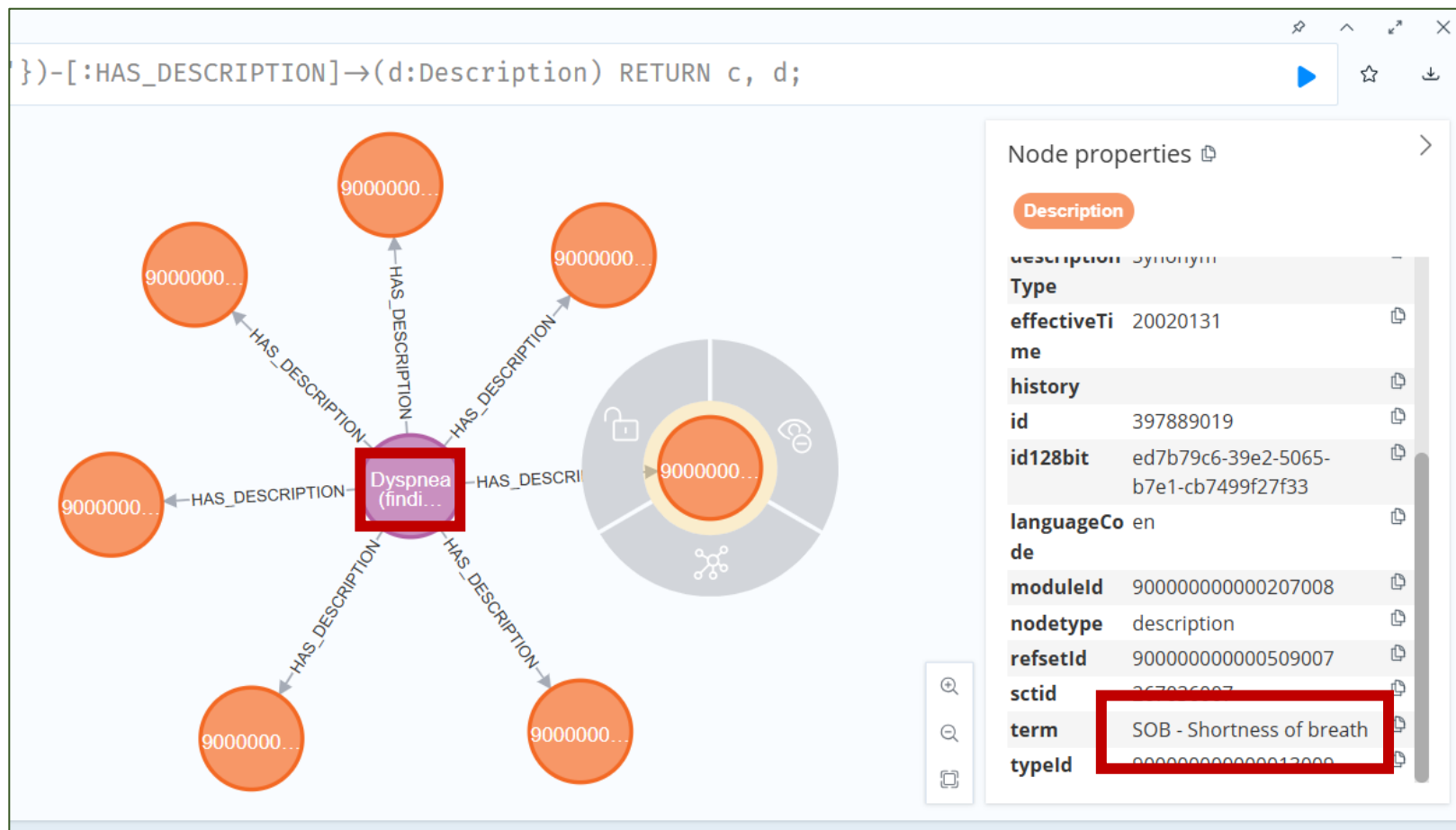


获取一个医学概念的相关属性。



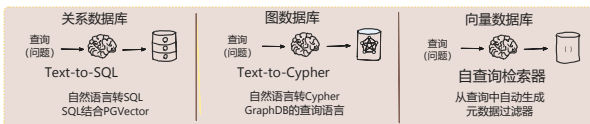
概念和属性

通过扩展概念的相关节点增强RAG准确率

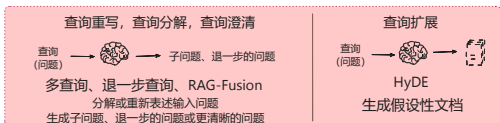


寻找突破口

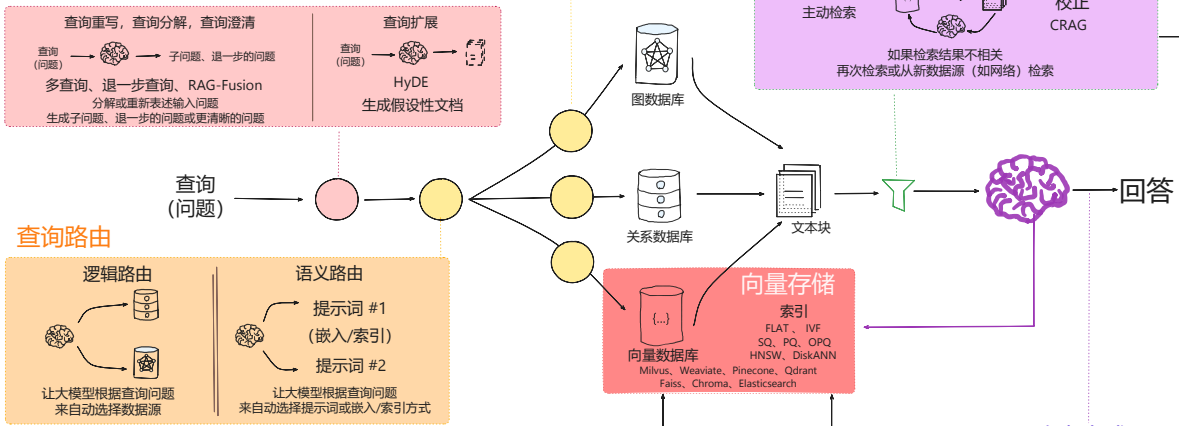
查询构建



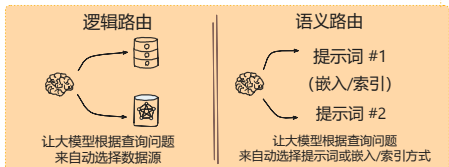
查询翻译



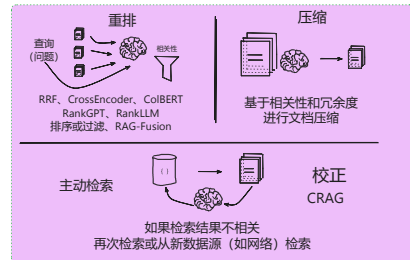
检索前处理



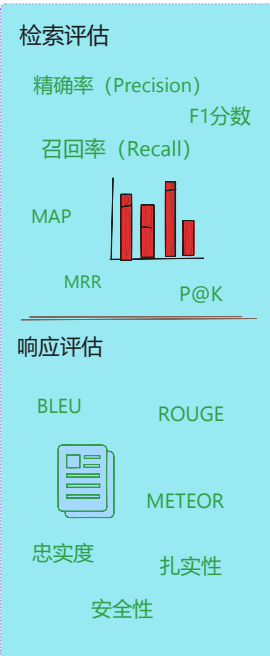
查询路由



检索后处理



系统评估



索引优化

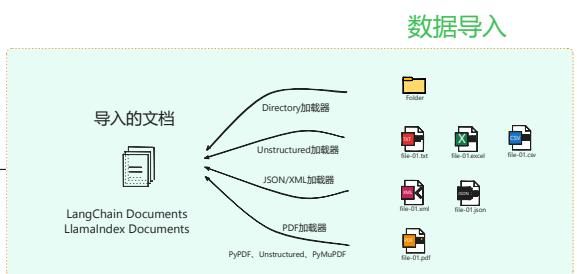
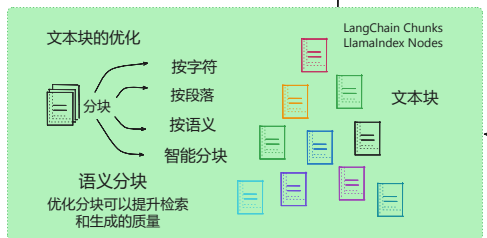


信息嵌入

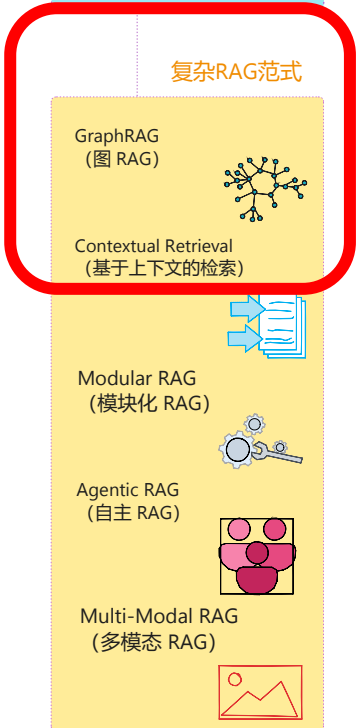


响应生成

文本分块



数据导入

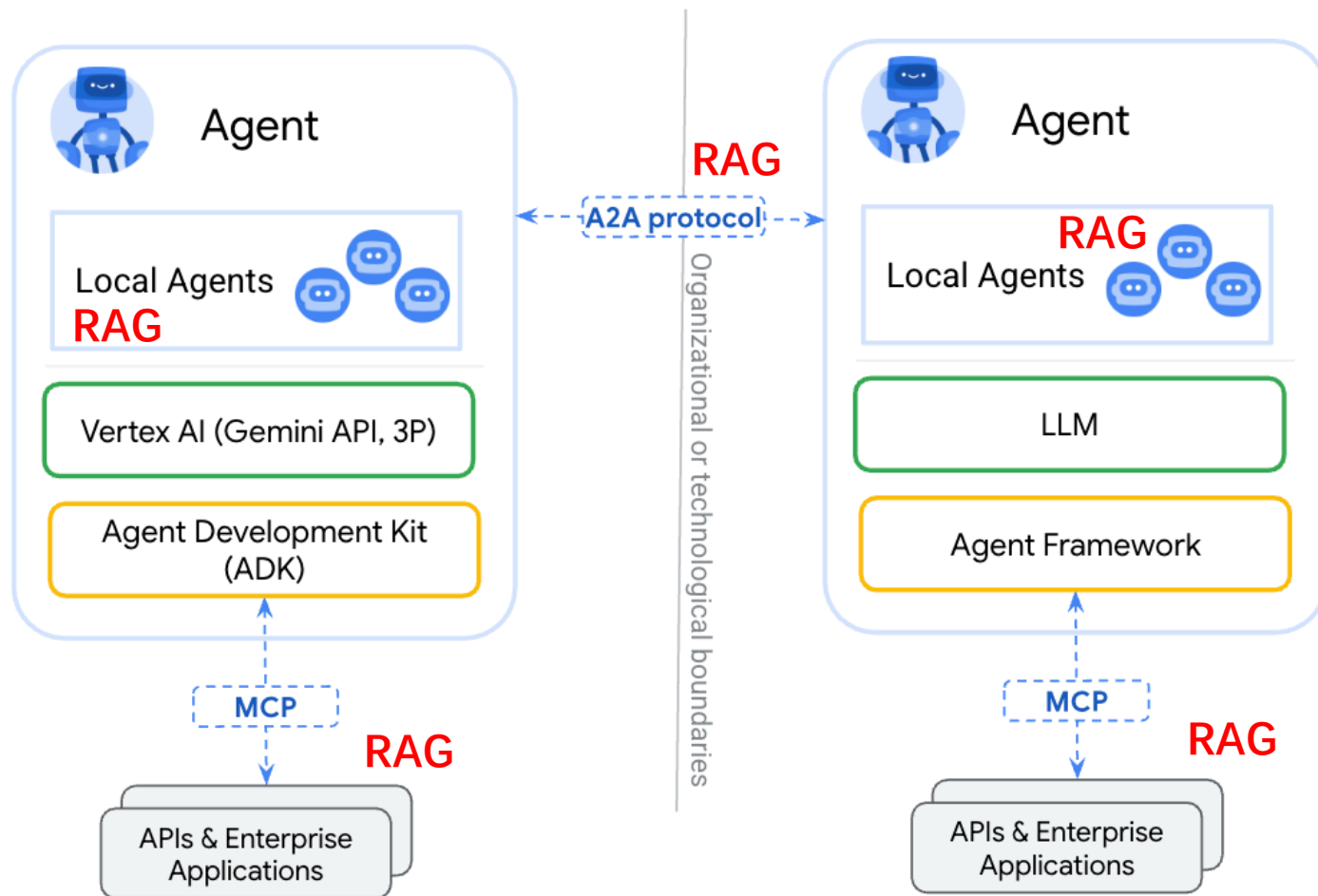


复杂RAG范式

06 MCP和A2A时代的RAG



- LLM: 是Agent毫无疑问的“大脑”，负责处理信息，推理，做决策。
- MCP: 负责模型与工具/资源的连接，是Agent的“手”，让Agent能够获取信息和执行操作。
- A2A: 负责Agent之间的通信，是Agent的“嘴”，让Agent能够相互交流、协作完成任务。



RAG 在这个生态中既是连接 LLM 与外部海量知识的纽带（通过 MCP 的“手”），也是多 Agent 协作时的信息载体（通过 A2A 的“嘴”）

📍 北京

QCon

全球软件开发大会

会议时间：4月10-12日

- 大模型赋能 AIOps
- 越挫越勇的大前端
- 云上业务架构演进
- 多模态大模型及应用
- 海外 AI 应用创新实践

📍 北京

AiCon

全球人工智能开发与应用大会

会议时间：6月27-28日

- 端侧智能
- AI Agent
- 多模态大模型
- 金融+大模型

📍 上海

QCon

全球软件开发大会

会议时间：10月23-25日

- AI Agent
- 大模型训练推理
- 端侧 AI
- 搜推深度融合
- 数智金融

4月

5月

6月

8月

10月

12月

📍 上海

AiCon

全球人工智能开发与应用大会

会议时间：5月23-24日

- 通用大模型
- AI Agent
- 垂直领域应用
- 模型可解释性

📍 深圳

AiCon

全球人工智能开发与应用大会

会议时间：8月22-23日

- 模型效率与部署
- 多模态大模型
- 大模型安全
- 智能硬件

📍 北京

AiCon

全球人工智能开发与应用大会

会议时间：12月19-20日

- 通用大模型
- 智能硬件
- LMOPs
- 具身智能

极客邦科技 2025 年会议规划

促进软件开发及相关领域知识与创新的传播



参会咨询



查看会议

THANKS

探索 AI 应用边界

Explore the limits of AI applications

AiCon

全球人工智能开发与应用大会